RMO(INDIA) – 1989(1)

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Is n = 7654321234567 a perfect square of some natural number? Justify your answer.
2. Let k be an integer having n digits. Let h be the sum of all the digits of k. Prove that k is divisible by 3 if and only if h is divisible by 3.
3. Is tan (1^0) a rational number or an irrational number? Justify your answer.
4. Prove any one of the following inequalities:
   i. Cauchy-Schwartz Inequality. For all real numbers $a_1, a_2, a_3, ..., a_n$ and $b_1, b_2, b_3, ..., b_n$ we have, $\left(a_1b_1 + a_2b_2 + ... + a_nb_n\right)^2 \le \left(a_1{}^2 + a_2{}^2 + ... + a_n{}^2\right)\left(b_1{}^2 + b_2{}^2 + ... + b_n{}^2\right)$
   ii. Cauchy inequality about generalized Arithmetic Mean and Geometric mean. For all positive real numbers $a_1, a_2, a_3, ..., a_n$; $\left(a_1a_2a_3...a_n\right)^{1/n} \le \dfrac{a_1 + a_2 + a_3 + ... + a_n}{n}$
5. a, b, c are natural numbers such that the sum of any two of them is greater than the third number. Prove that: $[1-(b-c)/a]^a[1-(c-a)/b]^b[1-(a-b)/c]^c \le 1$
6. Prove that $P(x) = a_nx^n + a_{n-1}x^{n-1} + ... + a_1x + a_0$ is a polynomial whose coefficients are integers. P(0) = 1989 and P(1) = 9891. Prove that P(x) = 0 possesses no integral solutions.
7. x = 1 and x = 2+I are solutions of $x^5 - 11x^4 | 46x^3 - 94x^2 + 93x - 35 = 0$. Find the other solutions
8. State the following theorems and prove any one of them:.
   i. Fermat's (little) theorem.
   ii. Euclid's algorithm for finding g.c.d of two integers.
   iii. Inclusion exclusion principle.
9. $a_1, a_2, a_3, ..., a_n$ is an ordered sequence of arbitrary integers. Prove that there exists an integer $k,\ 1 \le k \le n$, such that the sum of some k consecutive terms of the sequence is a multiple of n
10. State the following theorems and prove any one of them:
    i. Ceva's theorem about concurrency of lines.
    ii. Ptolemy's theorem about cyclic quadrilateral.
    iii. Simson's theorem about pedal line.
11. Two points A and B are on the same side of a line L. It is given that AB is neither parallel to L nor it is perpendicular to L. Construct a circle which passes through point A and B and which touches the line L. You are allowed to use an unmarked ruler and compass only.

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